MAHMUD YAHMAN MUHAMMED

Robotics & Mechatronics Engineering | PhD Applicant

Autonomous Exploration | Multi-Robot Systems | 3D Perception | Robot Learning

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RESEARCH PROFILE

Robotics and Mechatronics B.Eng. candidate interested in active perception, autonomous exploration, decentralized multi-robot coordination, and learning-based robotic control. Developed a ROS 2 3D frontier-detection package, trained a TD3 policy for simulated Panda manipulation, and implemented embedded robot-control systems. Seeking doctoral research on information-driven exploration, collaborative 3D mapping, and perception-aware autonomy; developing Active Viewpoint-Coupled Exploration (AVCE) as a longer-term research direction.

EDUCATION

B.Eng., Robotics & Mechatronics **Expected May 2027**

Belgorod State National Research University, Russia

- **Relevant coursework:** Robotics Systems, Control Systems, Embedded Systems, Signals & Systems, Engineering Programming, Applied Mechanics, CAD/CAM, Machine Learning Theory and Algorithms.

SELECTED RESEARCH & ROBOTICS PROJECTS

Frontier Detector 3D GitHub

Independent Robotics Research Project | ROS 2, 3D Perception & Exploration

Technologies: ROS 2, Python, C++, RViz 2, sparse voxel mapping

- **Contribution:** Built a ROS 2 3D frontier detector using sparse voxel mapping, 6-neighbor frontier identification, 26-connected BFS clustering, information-gain ranking, and goal publication for exploration.
- **Verification:** Passed 26 automated tests of frontier detection, clustering, and goal ranking; demonstrated markers and ranked goals in a synthetic 3D scene, with a configured 2 Hz update rate.
- **Benchmark status:** A controlled comparison with an existing 3D frontier detector on matched voxel scenes (frontier agreement, runtime, memory) is proposed but has not yet been performed.

Deep Reinforcement Learning for Panda Door Manipulation GitHub

Independent Robotics Project | TD3, MuJoCo / robosuite

Technologies: Python, PyTorch, TD3, MuJoCo, robosuite, CUDA

- **Implementation:** Built a pure-TD3 actor–critic training pipeline for a simulated Franka Panda door task, using replay, checkpoints, 64-dimensional observations, and 8-dimensional joint-velocity/gripper actions.
- **Evaluation:** Best checkpoint achieved 40/50 successful door openings (80%) in simulation; an earlier checkpoint reached 5/10 (50%) in a separate evaluation, not a controlled head-to-head baseline.
- **Failure analysis:** Identified reward-induced shortcut behavior: the original policy opened the door before valid grasp alignment in 21/30 trials. Designed a staged alignment curriculum; reliable two-pad handle contact remains unresolved.

Vision-Guided TD3 vs. Pretrained VLA for Robotic Manipulation GitHub **Research design | In progress**

Independent Benchmark Project

Technologies: Python, PyTorch, CNNs, TD3, VLA policies, LIBERO, MuJoCo

- **Study design:** Designed a matched-task benchmark comparing a CNN-assisted visual TD3 controller against a compatible pretrained vision-language-action policy on a simulated Franka Panda pick-and-place task.
- **Evaluation plan:** Defined planned comparisons of task success, visual robustness, inference latency, and compute requirements. Policy compatibility, implementation, and controlled comparative experiments remain in progress.

SELENE: Autonomous Two-Wheeled Rover GitHub **ESP32-S3 | micro-ROS**

Technologies: C/C++, ROS 2, micro-ROS, quadrature encoders, Wokwi simulation

- **Implementation:** Developed an ESP32-S3 rover firmware architecture with a finite-state-machine controller, motor/encoder modules, differential-drive odometry, and ROS 2 communication through micro-ROS.
- **Design considerations:** Kept critical rover states on-board to handle micro-ROS agent disconnections and limited outgoing topics to reduce communication overhead; used Wokwi to test embedded behavior.

PIVOT: Self-Balancing Two-Wheeled Robot GitHub

Technologies: ROS 2, Gazebo, URDF/Xacro, Onshape CAD, Python/C++, RViz, PID

- **Implementation:** Developed a parametric CAD-to-URDF/Xacro robot description with consistent ROS frames, visual/collision geometry, and inertial properties; also developed balancing-control experiments in Gazebo.
- **Next steps:** Final-year thesis project direction; integrating SLAM and Nav2 for autonomous navigation, completing the simulation phase before developing the physical robot for my final-year thesis, "**Development of a System for Stabilizing and Controlling the Movement of a Two-Wheeled Balancing Robot – PIVOT**," with the ultimate goal of achieving a complete **sim-to-real** pipeline.

Heterogeneous Multi-Robot Autonomy Architecture GitHub

- Designed a modular ROS 2 / Gazebo autonomy architecture for heterogeneous UGV/UAV teams. Documented a working UGV simulation bridge and sensor/command interfaces; collaborative fusion and multi-robot trials remain in progress.

PROFESSIONAL EXPERIENCE

AI & Robotics Intern, Flolabs **Jan 2026 - Present**

CAIPO IoT Team; Robotics & Humanoid Team

- Developed a wearable-device LED state-management library with 17 states, priority logic, base-state handling, and a Python GUI companion for ESP32-S3.
- Contribute to humanoid-robot sensor integration, control validation, subsystem testing, and experiment documentation.
- Documenting experiments and manufacturing structured development logs for performance evaluation, Create bills of materials and propose purchase and logistics route for essential build materials.

AI Research Intern (Game Development), Akiyam Solutions **Feb - Jun 2026**

- Investigated machine-learning applications for adaptive gameplay, intelligent NPC behavior, and agent decision-making.
- Analyzed performance and scalability of AI-driven gameplay mechanics.

Undergraduate Research Assistant **Sep 2025 - Present**

Robotics Laboratory | Belgorod State National Research University

- Conduct faculty-supervised robotics research and hands-on prototyping, applying ROS 1/ROS 2, control theory, and navigation concepts.
- Use lab access to sensors, electronic components, and 3D printing to explore hardware feasibility and prototype robot systems.
- Refined the SELENE rover concept through laboratory feedback and feasibility checks; gained exposure to a humanoid robotics platform.

TECHNICAL SKILLS

Programming & infrastructure: Python, C/C++, Linux (Ubuntu), Git/GitHub, ROS 2, micro-ROS
Robotics & simulation: MuJoCo, robosuite, Gazebo, URDF/Xacro, RViz 2, PyBullet, CoppeliaSim, Webots
Learning & perception: PyTorch, TD3, OpenCV, Open3D, 3D mapping, frontier exploration, SLAM, A*/RRT; PPO fundamentals
CAD & embedded: Onshape, T-FLEX CAD, FreeCAD, KOMPAS-3D, SolidWorks; ESP32-S3, Arduino, MSP430, KiCad, motor drivers, encoders, LiDAR/ToF sensors

AWARDS & HONORS

Aurora Scholarship | Merit-based scholarship recipient, Aurora Robotics
BEA Scholarship | Nigerian Federal Government, October 2022
KFUPM Inbound Summer Research Award | KFUPM, 2026
Honor Student / Face of the Faculty | Kursk State University, May 2023
Russian Language Olympiad Winner | Kursk State University, January 2023

LEADERSHIP & ACTIVITIES

President, Nigerian Community in Belgorod **2026 - Present**
Public Relations Officer, ANSSIR Belgorod **2024 - 2026**
Head of Media Team, ANSSIR **2025 - 2026**